

No Undeclared Relation: Bridge Completion and No-Free-Transfer in Foundational Inference

Subtitle: Declared Bridges, Target Distinctions, and Constructive Bridge Completion

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Abstract

Foundational arguments often move from one closure context to another. Examples include entropy to geometry, recoverability to causality, positive information geometry to Lorentzian structure, local compatibility to global objecthood, residual structure to dynamics, and prediction to explanation. This paper argues that every such arrow is a relation claim. In a relation-first framework, a relation claim is not valid unless the relation itself is declared, typed, and assigned preservation conditions.

The master principle is:

No declared relation, no legitimate relational inference.

The operative cross-context form is therefore not a bare dyad $X \rightarrow Y$, but a triad:

$$X - \mathfrak{B}_{X \rightarrow Y} - Y.$$

The cross-context corollary is the No-Free-Transfer theorem. In particular, every apparent dyadic transfer $X \rightarrow Y$ is underdeclared until the mediating bridge relation is supplied, making the operative structure $X - \mathfrak{B}_{X \rightarrow Y} - Y$. Let X and Y be closure contexts with task-visible distinction relations D_X and D_Y . A cross-context inference

$$X \Rightarrow Y$$

is valid only if the Y -distinctions claimed by the conclusion are recoverable from X -distinctions under declared bridge data, or if they are formal boundary/definitional consequences common to the framework. Otherwise the inference is a no-free-transfer failure.

The paper also extracts the programme's no-go network: boundary, quotient, triadic, chart, objecthood, learning, residual, and external-transfer no-gos are treated as projections of the same relation-first constraint. This makes the paper more than a bridge taxonomy; it shows how the theorem machinery generated across the companion papers coheres under one primitive discipline.

The constructive method is:

audit	→	bridge-type classification	→	candidate bridge completion	→	calibration.
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Given a claim $X \Rightarrow Y$, the bridge audit chart asks five questions: what is the source context, what is the target context, what target distinctions are claimed, are those target distinctions already encoded in the source, and what bridge data would preserve them if not. The audit does not refute the claim. It makes the claimed relation inspectable.

The paper then applies the method to representative foundations transfers and gives four worked bridge-completion attempts. The strongest quantitative result is the persistence-margin trace-stability theorem: if a path trace's threshold policy remains inside persistence support with perturbation smaller than the support margin, then the trace remains in the proper organization layer. Failure at or beyond the margin identifies the next learning move: recalibration, recharting, or a new threshold policy.

0. Scope and relationship to the companion programme

This paper is not a new constructional theory. It uses the companion theories as already declared structures:

1. **Theory of Organization** supplies organization, zero-boundary, proper interior, persistence, and evaluator machinery.
2. **Charted Organization Theory** supplies observer-charts, atlases, transitions, objectivity, and objecthood.
3. **Recursive Chart Learning Theory** supplies path traces, branch fields, evaluators, admission, calibration, and trace compression.
4. **Triadic Closure and Recursive Residual Theory** supplies boundary-mediation-visibility, closure residuals, and triadic mediation.
5. **Cross-Closure Instantiation Theorems** supplies bridge declarations, schema-to-chart induction, residual-to-learning pressure, persistence-compatible traces, three-chart mediation residuals, and operation-trace-closure preorders.
6. The quotient-admissibility discipline is used locally as a preservation principle: preservation claims require declared preservation data. The external-facing quotient paper develops this as a bounded application, but the present paper states the needed bridge discipline directly.

This paper merges two functions:

no-free-transfer theorem

and:

constructive bridge-completion method.

The theorem supplies the negative discipline:

do not transfer target distinctions for free.

The no-go network section shows that this discipline is already active across the companion papers: boundary no-gos, quotient no-gos, chart no-gos, objecthood no-gos, learning no-gos, residual no-gos, and external bridge no-gos are all projections of the same No-Undeclared-Relation principle.

The constructive method supplies the positive discipline:

identify the missing bridge type and search for solved bridge templates.

0.1 Master principle: no undeclared relation

The bridge in this paper is not an optional explanatory ornament. It is the relation claimed by a cross-context inference. If an argument writes:

$$X \Rightarrow Y,$$

then it asserts that a relation carries some structure from X to Y . In a relation-first framework, that relation cannot remain implicit. It must be declared, typed, and assigned preservation conditions.

Thus No-Free-Transfer is a corollary of the stronger principle:

No-Undeclared-Relation.

A cross-context arrow that lacks bridge data is not merely incomplete. It is missing the relation that would make the inference structurally meaningful.

The formal theorem below is elementary by design. Its force is methodological: every foundations arrow must identify what relation performs the transfer and what target distinctions that relation preserves.

0.2 Adjacent methodological frameworks

This paper is adjacent to several existing methodological traditions.

- In **analogical reasoning and model transfer**, a source model supplies candidate structure for a target model, but the analogy is not proof. The bridge-template discipline here is similar: a solved bridge supplies a candidate completion only after target adequacy is recalibrated.

- In **negative-results and null-results methodology**, failed attempts can be evidentially valuable when they identify which assumptions fail. The bridge calibration theorem treats failed bridge completion as learning because it localizes the missing preservation component.
- In **Pearl’s causal hierarchy**, associational, interventional, and counterfactual claims occupy different closure contexts. The hierarchy is a canonical example of no-free-transfer: counterfactual distinctions do not transfer freely from associational data.
- In **robustness analysis**, a result is more credible when it survives perturbations across models or assumptions. The persistence-margin theorem below is a formal robustness statement for path traces under bounded threshold perturbation.
- In **category-theoretic language**, bridge declarations resemble structure-preserving maps, bridge composition resembles morphism composition, and bridge adequacy resembles preservation of specified structure. No categorical machinery is assumed here, but the analogy clarifies why bridge composition requires compatibility data.

These adjacent frameworks are comparison points, not premises. The paper’s internal claim is relation-first: an inference is a relation claim, and undeclared relations do not transfer distinctions.

0.3 Diagnostic studies versus worked theorem attempts

Sections 13–18 are **diagnostic case studies**. They apply the audit chart to familiar foundations transfers and identify required bridge types.

Section 21 contains **worked bridge-completion attempts**. These go beyond diagnosis by proving success or failure theorems under declared bridge data. The persistence-margin theorem in §21.4 is the main quantitative cross-paper result of the paper.

0.4 Careful-literature versus compressed-slogan convention

When the paper audits a mature field, it targets compressed transfer slogans rather than careful technical literature. For example, the audit of entropy $\Rightarrow$ geometry is not a charge that holographic-entanglement researchers fail to declare bridge data. Responsible work in that field does declare dictionaries, code subspaces, area relations, region classes, reconstruction maps, and dynamical assumptions. The audit instead identifies what must be present whenever the slogan is used.

1. Closure contexts and target distinctions

Definition 1.1 — Closure context

A **closure context** is a structured domain X equipped with:

$$\mathfrak{X} = (X, \mathcal{T}_X, D_X, \mathcal{A}_X, \text{Adm}_X),$$

where:

- X is a carrier, chart, atlas, organization layer, trace space, residual space, or other structured domain;
- $\mathcal{T}_X$ is a declared task family;

- $D_X \subseteq X \times X$ is the task-visible distinction relation;
- $\mathcal{A}_X$ is the admissible structure through which X -distinctions are read;
- Adm_X is the admissibility condition for X -claims.

If no task family is declared, D_X is undefined and no nontrivial preservation claim about X -distinctions is meaningful.

Definition 1.2 — Target distinction

Let:

$$\mathfrak{Y} = (Y, \mathcal{T}_Y, D_Y, \mathcal{A}_Y, \text{Adm}_Y)$$

be a closure context. A **target distinction** is an ordered pair:

$$(y, y') \in D_Y.$$

A cross-context claim from X to Y is a claim that some family of Y -target distinctions is determined, preserved, recovered, generated, or constrained by X .

Definition 1.3 — Source indistinguishability

Given a closure context X , define the source indistinguishability relation:

$$x \sim_X x'$$

when no declared X -task distinguishes x from x' .

Equivalently, $x \sim_X x'$ iff x and x' agree on all declared task-visible X -distinctions.

Definition 1.4 — Cross-context inference

A **cross-context inference** is a claim of the form:

$$\mathfrak{X} \Rightarrow \mathfrak{Y}$$

asserting that structure in X determines, recovers, preserves, or constrains structure in Y .

A cross-context inference is **nontrivial** if at least one target distinction in D_Y is not definitional in Y and not a boundary condition common to the framework.

2. Bridge declarations

2.0 Source–bridge–target triad

Every cross-context bridge claim has the form:

$$X \xrightarrow{\mathfrak{B}_{X \rightarrow Y}} Y.$$

This is not a dyad X, Y . It is a triad:

source context — declared bridge relation — target context.

The source context supplies what is available.

The target context supplies what is claimed.

The bridge relation supplies the preservation, comparison, recovery, or translation structure by which the target claim can be read from the source.

Thus a bare arrow

$$X \rightarrow Y$$

is underdeclared unless the middle term is supplied. The bridge is not decorative notation added to an already meaningful transfer. The bridge is the relation that makes the transfer a relation-first claim.

This is the cross-context instance of the programme’s triadic discipline:

boundary/source + mediation/bridge + visibility/target-readout.

The source and target alone do not determine what is preserved, what is erased, what is compared, or what counts as successful recovery. The third term is therefore not optional.

Proposition 2.0.1 — Dyadic transfer is underdeclared

A claimed transfer $X \rightarrow Y$ that names only source and target, but supplies no bridge relation $\mathfrak{B}_{X \rightarrow Y}$, is not yet a valid cross-context inference.

Proof A nontrivial target claim requires recovery of some target distinction family $D'_Y \subseteq D_Y$. Recovery is defined only relative to bridge data:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}}(D_X).$$

If no bridge relation $\mathfrak{B}$ is declared, then $\text{Rec}_{\mathfrak{B}}(D_X)$ is undefined. Hence the transfer cannot establish D'_Y . $\square$

Remark 2.0.2 — Binary relations and triadic inference

A bridge may be represented set-theoretically as a binary relation, map, functor, comparison, or transport. This does not make the inference dyadic. The inferential structure remains triadic because the relation is being used to mediate between a source context and a target context:

$$(X, \mathfrak{B}_{X \rightarrow Y}, Y).$$

The third component is not a third object. It is the declared relation-mode that makes the source and target comparable.

Remark 2.0.3 — Relation to the triadic mediation inequality

The source–bridge–target triad is the primitive relation-form of the triadic mediation principle. It is prior to any metric or distance assignment.

Once a bridge carries path-cost, residual, accessibility, or comparison structure, the triadic form may be read through a generalized mediation inequality:

$$\Delta(X, Z) \preceq \Delta(X, Y) \otimes \Delta(Y, Z).$$

When the value structure is the ordinary nonnegative real line with $\otimes = +$ and $\preceq = \leq$, this becomes the usual triangle inequality:

$$d(X, Z) \leq d(X, Y) + d(Y, Z).$$

Thus the triangle inequality does not replace No-Undeclared-Relation. Rather, it is a metric realization of the same deeper point: a direct relation is meaningful only inside a declared mediation structure. No-Undeclared-Relation is the primitive constraint; the triangle inequality is one path-cost shadow of that constraint.

Definition 2.1 — Bridge declaration

A **bridge declaration** from $\mathfrak{X}$ to $\mathfrak{Y}$ is declared data:

$$\mathfrak{B}_{X \rightarrow Y}$$

specifying how X -structure is read as, transported to, compared with, or recovered as Y -structure.

Bridge declarations may include:

1. a chart or bridge chart;
2. comparison maps between value spaces;
3. faithful readouts;
4. quotient maps;
5. residual readouts;

6. threshold transport policies;
7. persistence compatibility conditions;
8. task-loss bridges;
9. transition functors;
10. calibration or evaluator compatibility;
11. admissibility relations;
12. recovery maps;
13. gluing or global-section data;
14. action/coupling/update rules.

A bridge declaration is part of the theorem data, not a rhetorical supplement.

Definition 2.2 — Bridge recovery of distinctions

A bridge declaration $\mathfrak{B}_{X \rightarrow Y}$ **recovers** a target distinction:

$$(y, y') \in D_Y$$

from X if there are X -distinctions or X -admissible data sufficient under $\mathfrak{B}_{X \rightarrow Y}$ to determine that y and y' remain distinguished in Y .

Write:

$$(y, y') \in \text{Rec}_{\mathfrak{B}}(D_X)$$

when the Y -distinction (y, y') is bridge-recoverable from D_X .

Definition 2.3 — Adequate bridge

A bridge declaration $\mathfrak{B}_{X \rightarrow Y}$ is **adequate for** $D'_Y \subseteq D_Y$ if:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}}(D_X).$$

That is, every claimed target distinction is recoverable from source distinctions under the bridge.

Definition 2.4 — Boundary/definitional distinction

A target distinction is **boundary/definitional** if it follows solely from:

1. initial/terminal boundary discipline;
2. excluded-extrema definitions;
3. formal identity/equality inside the declared context;
4. algebraic rules explicitly defining the context;
5. proper-interior conditions already built into the context.

Boundary/definitional distinctions do not require cross-context bridge data because they are not transferred from X to Y ; they are already part of the declared target context.

2.1 Triadic declaration of relation

Companion-paper note. The face-corollaries of this section are developed in *No Undeclared Boundary* and *No Undeclared Observer*. The closure-atlas navigation consequences are developed structurally in *Triadic Substitution, Coupling, and Closure Atlases* and operationally in *Recursive Bridge Calibration*.

The source–bridge–target triad is the cross-context expression of a more general declaration law.

A declared relation is not merely a named arrow. It requires a closure-complete triadic role structure:

$$R_{\text{declared}} \Rightarrow (Z_R, R_R, O_R).$$

Where:

- Z_R is the boundary / distinction / admissibility face;
- R_R is the mediation / bridge / transition face;
- O_R is the observer / readout / evaluator face.

Thus No Undeclared Relation has the expanded interpretation:

No relation whose boundary, mediation, or readout face is undeclared.

This does not introduce new primitives over relation. Boundary and observer/readout are necessary faces of declared relation.

Proposition 2.1.1 — Boundary and observer failures are NUR failures

If a relation claim relies on an undeclared boundary or an undeclared observer/readout, then it violates No Undeclared Relation.

Proof By triadic declaration, a relation is declared only when its boundary, mediation, and observer/readout faces are declared. If the boundary face is missing, the relation’s distinction or admissibility condition is underdeclared. If the observer/readout face is missing, the relation’s visibility or evaluation condition is underdeclared. In either case the relation itself is underdeclared. $\square$

Remark 2.1.2 — Relation face and mediation

The mediation face is not a fourth principle. It is the relation face governed directly by NUR. No Undeclared Boundary and No Undeclared Observer are therefore face-corollaries of NUR, not sibling master axioms.

3. No-Free-Transfer as cross-context corollary

Theorem 3.1 — No-Free-Transfer

Let:

$$\mathfrak{X} = (X, \mathcal{T}_X, D_X, \mathcal{A}_X, \text{Adm}_X)$$

and:

$$\mathfrak{Y} = (Y, \mathcal{T}_Y, D_Y, \mathcal{A}_Y, \text{Adm}_Y)$$

be closure contexts. Let $D'_Y \subseteq D_Y$ be a family of non-boundary, non-definitional target distinctions claimed to follow from X .

If no adequate bridge declaration:

$$\mathfrak{B}_{X \rightarrow Y}$$

is supplied such that:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}}(D_X),$$

then the inference:

$$\mathfrak{X} \Rightarrow \mathfrak{Y}$$

does not validly establish D'_Y .

Equivalently:

$$\boxed{D'_Y \not\subseteq \text{Rec}_{\mathfrak{B}}(D_X) \quad \Rightarrow \quad X \not\Rightarrow_{D'_Y} Y.}$$

Proof

A non-boundary target distinction $(y, y') \in D'_Y$ is not guaranteed by the target context's definitions or boundary discipline. To establish it from X , the inference must show that X -structure distinguishes the corresponding target alternatives after translation into Y -terms.

If no adequate bridge is supplied, then by assumption there is at least one $(y, y') \in D'_Y$ such that:

$$(y, y') \notin \text{Rec}_{\mathfrak{B}}(D_X).$$

Thus no declared source distinction or source-admissible datum preserves or recovers that target distinction. Therefore the claimed inference fails to establish all target distinctions in D'_Y . Hence:

$$X \not\Rightarrow_{D'_Y} Y.$$

□

Remark 3.1.1 — Elementary proof, load-bearing discipline

The proof of Theorem 3.1 is intentionally elementary. The theorem is not presented as a difficult mathematical discovery. It is the formal expression of the No-Undeclared-Relation principle: a cross-context arrow has no structural force unless the relation performing the transfer is declared and shown to preserve the target distinctions.

Corollary 3.2 — Vocabulary alignment is not transfer

If X and Y share vocabulary, analogous notation, similar diagrams, or structurally similar labels, but no adequate bridge declaration is supplied, then no nontrivial cross-context inference follows.

Proof

Vocabulary alignment alone does not define $\text{Rec}_{\mathfrak{B}}(D_X)$. Without bridge data, no inclusion:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}}(D_X)$$

can be established. Theorem 3.1 applies. □

Corollary 3.3 — Boundary exception

If a target statement is purely boundary/definitional, it may hold without bridge data.

This does not contradict Theorem 3.1 because Theorem 3.1 concerns non-boundary, non-definitional target distinctions.

4. Relation to task-visible quotient admissibility

The No-Free-Transfer theorem is the cross-context analogue of the quotient admissibility criterion.

In the quotient-admissibility discipline, a quotient:

$$q : X \rightarrow Q$$

is task-visible-admissible when:

$$\boxed{\sim_q \subseteq \sim_{\mathcal{T}}^R .}$$

That is, the quotient may identify only distinctions invisible to the declared task and structure type.

The present theorem says the analogous thing for cross-context inference:

$$\boxed{D'_Y \subseteq \text{Rec}_{\mathfrak{B}}(D_X) .}$$

That is, the target distinctions claimed in Y must be recoverable from X under declared bridge data.

Thus:

$$\boxed{\text{quotient admissibility} = \text{within-context no-free-erasure};}$$

$$\boxed{\text{bridge admissibility} = \text{cross-context no-free-transfer}.}$$

5. Bridge adequacy, minimality, and strength

Definition 5.1 — Bridge strength preorder

Let $\mathfrak{B}_1, \mathfrak{B}_2$ be bridge declarations from X to Y . Define:

$$\mathfrak{B}_1 \preceq \mathfrak{B}_2$$

iff:

$$\text{Rec}_{\mathfrak{B}_1}(D_X) \subseteq \text{Rec}_{\mathfrak{B}_2}(D_X).$$

Thus $\mathfrak{B}_2$ is at least as strong as $\mathfrak{B}_1$ if it recovers every target distinction recovered by $\mathfrak{B}_1$.

This bridge-strength preorder is the cross-context analogue of the resource/preorder discipline used in Paper B. Stronger bridge data recover at least as many target distinctions, just as a more informative representation preserves at least as many task-visible distinctions.

Proposition 5.2 — Bridge monotonicity

If:

$$\mathfrak{B}_1 \preceq \mathfrak{B}_2$$

and $\mathfrak{B}_1$ is adequate for D'_Y , then $\mathfrak{B}_2$ is adequate for D'_Y .

Proof

Adequacy of $\mathfrak{B}_1$ gives:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}_1}(D_X).$$

Bridge strength gives:

$$\text{Rec}_{\mathfrak{B}_1}(D_X) \subseteq \text{Rec}_{\mathfrak{B}_2}(D_X).$$

Therefore:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}_2}(D_X).$$

So $\mathfrak{B}_2$ is adequate. $\square$

Definition 5.3 — Minimal bridge

A bridge declaration $\mathfrak{B}$ adequate for D'_Y is **minimal** if no proper subdeclaration $\mathfrak{B}' \subsetneq \mathfrak{B}$ remains adequate for D'_Y .

Minimality is relative to a declared notion of subdeclaration. It may be by component removal, resource preorder, definitional strength, or preservation strength.

Proposition 5.4 — Missing component no-go for minimal bridges

If $\mathfrak{B}$ is minimal for D'_Y , then every component of $\mathfrak{B}$ is load-bearing: removing it destroys adequacy.

Proof

By definition of minimality, no proper subdeclaration is adequate. Removing any component produces a proper subdeclaration. Therefore the resulting bridge is inadequate. $\square$

6. Bridge composition and incompatibility

Definition 6.1 — Composable bridges

Let:

$$\mathfrak{B}_{X \rightarrow Y}$$

and:

$$\mathfrak{B}_{Y \rightarrow Z}$$

be bridge declarations. They are **composable** if the target distinctions recovered in Y by $\mathfrak{B}_{X \rightarrow Y}$ include all source distinctions required by $\mathfrak{B}_{Y \rightarrow Z}$.

Equivalently, if D_Y^{req} denotes the Y -distinctions needed to activate $\mathfrak{B}_{Y \rightarrow Z}$, then:

$$D_Y^{\text{req}} \subseteq \text{Rec}_{\mathfrak{B}_{X \rightarrow Y}}(D_X).$$

Definition 6.2 — Composite bridge

When $\mathfrak{B}_{X \rightarrow Y}$ and $\mathfrak{B}_{Y \rightarrow Z}$ are composable, their composite is:

$$\mathfrak{B}_{X \rightarrow Z} = \mathfrak{B}_{Y \rightarrow Z} \circ \mathfrak{B}_{X \rightarrow Y}.$$

Its recovered distinctions are:

$$\text{Rec}_{\mathfrak{B}_{X \rightarrow Z}}(D_X) = \text{Rec}_{\mathfrak{B}_{Y \rightarrow Z}}\left(\text{Rec}_{\mathfrak{B}_{X \rightarrow Y}}(D_X)\right).$$

Theorem 6.3 — Bridge composition theorem

If:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}_{X \rightarrow Y}}(D_X)$$

and:

$$D'_Z \subseteq \text{Rec}_{\mathfrak{B}_{Y \rightarrow Z}}(D'_Y),$$

then:

$$D'_Z \subseteq \text{Rec}_{\mathfrak{B}_{X \rightarrow Z}}(D_X).$$

Proof

By the first inclusion, the relevant Y -distinctions are recoverable from X . By the second inclusion, the target Z -distinctions are recoverable from those Y -distinctions. By definition of composite bridge recovery:

$$\text{Rec}_{\mathfrak{B}_{X \rightarrow Z}}(D_X) = \text{Rec}_{\mathfrak{B}_{Y \rightarrow Z}}\left(\text{Rec}_{\mathfrak{B}_{X \rightarrow Y}}(D_X)\right).$$

Therefore D'_Z is contained in the composite recovery set. $\square$

Definition 6.4 — Bridge incompatibility

Two bridges:

$$\mathfrak{B}_{X \rightarrow Y}, \quad \mathfrak{B}_{Y \rightarrow Z}$$

are **incompatible** if $\mathfrak{B}_{X \rightarrow Y}$ fails to recover at least one distinction required by $\mathfrak{B}_{Y \rightarrow Z}$.

Corollary 6.5 — Bridge incompatibility blocks composition

If $\mathfrak{B}_{X \rightarrow Y}$ and $\mathfrak{B}_{Y \rightarrow Z}$ are incompatible, then the composite inference:

$$X \Rightarrow Z$$

cannot be validly established through those bridges.

Proof

By incompatibility, a required Y -distinction is absent from:

$$\text{Rec}_{\mathfrak{B}_{X \rightarrow Y}}(D_X).$$

Thus the hypothesis of Theorem 6.3 fails. Therefore no valid composite bridge inference follows. $\square$

7. The bridge audit chart

Definition 7.1 — Transfer claim

A **transfer claim** is an asserted inference:

$$X \Rightarrow Y$$

from a source closure context X to a target closure context Y .

The claim may be explicit or compressed in prose. Examples include:

$$\text{entropy} \Rightarrow \text{geometry},$$

recoverability $\Rightarrow$ causality,

positive information metric $\Rightarrow$ Lorentzian metric,

local compatibility $\Rightarrow$ global objecthood.

Definition 7.2 — Bridge audit chart

The **bridge audit chart** for a transfer claim $X \Rightarrow Y$ is the five-question relation-map:

$$\mathcal{C}_{\text{audit}}(X \Rightarrow Y) = (Q_1, Q_2, Q_3, Q_4, Q_5),$$

where:

Q_1 = source context identification,

Q_2 = target context identification,

Q_3 = target distinction identification,

Q_4 = source encoding test,

Q_5 = missing bridge classification.

The audit chart is itself a chart in the programme's sense: it is a relation-map that makes the transfer claim locally legible by separating source, target, target distinctions, source encoding, and required bridge data.

Proposition 7.3 — Audit chart as bridge declaration precursor

A bridge audit chart does not itself supply bridge data. It supplies the chart in which bridge data can be named.

Proof

The five audit questions identify the source context, target context, and target distinctions. They do not construct a recovery, readout, comparison map, chartability map, faithful loss, or transport

relation. Therefore the audit chart is prior to a bridge declaration. It makes the missing bridge visible but does not fill it. $\square$

8. Bridge as relation, path, trace, resolution, and closure bound

A bridge is not an external explanatory ornament. It is relation-first.

Definition 8.1 — Bridge relation

A **bridge relation** from X to Y is a relation-class:

$$\mathfrak{B}_{X \rightarrow Y}$$

whose task is to preserve or recover declared Y -distinctions from declared X -distinctions.

Definition 8.2 — Bridge path

A **bridge path** is a directed sequence of bridge relations:

$$X = X_0 \xrightarrow{\mathfrak{B}_0} X_1 \xrightarrow{\mathfrak{B}_1} \dots \xrightarrow{\mathfrak{B}_{n-1}} X_n = Y.$$

Definition 8.3 — Bridge trace

A **bridge trace** is an admitted bridge path together with its declared preservation data, evaluator, or adequacy test.

Thus a bridge trace records not only that X was related to Y , but how the relation preserved the target distinctions.

Definition 8.4 — Bridge resolution

A bridge resolves a transfer claim when the target distinctions D'_Y are recovered:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}}(D_X).$$

Definition 8.5 — Closure bound of a bridge

The **closure bound** of a bridge is the limit of what the bridge can preserve. Formally, it is the maximal target distinction family $D_Y^{\max} \subseteq D_Y$ such that:

$$D_Y^{\max} \subseteq \text{Rec}_{\mathfrak{B}}(D_X).$$

A bridge may resolve one target family and fail another.

Proposition 8.6 — Bridge relation principle

Every legitimate bridge declaration is a relation-class, path, trace, resolution, or closure bound, depending on the level of description.

Proof

A bridge declaration relates source distinctions to target distinctions. As a relation, it is a bridge relation. If the relation is decomposed across intermediate contexts, it is a bridge path. If admitted and recorded with preservation data, it is a bridge trace. If adequate for a target distinction family, it is a bridge resolution. Its maximal preservation scope is its closure bound. These are not separate primitives; they are faces of one bridge structure. $\square$

9. Bridge template transfer

The constructive method begins when a solved case and an underdeclared case share a bridge profile.

Definition 9.1 — Bridge profile

A **bridge profile** is a tuple:

$$\Pi(\mathfrak{B}) = (S, T, D_T, \mathcal{R}, \mathcal{A}),$$

where:

- S is the source type;
- T is the target type;
- D_T is the target distinction family;
- $\mathcal{R}$ is the recovery/readout/comparison form;
- $\mathcal{A}$ is the adequacy condition.

Definition 9.2 — Solved bridge template

A **solved bridge template** is a bridge declaration $\mathfrak{B}^*$ whose adequacy for its target distinctions has been proved or accepted in a mature theory.

Examples include:

- causal-order plus volume in causal set continuum approximation;
- intervention structure in structural causal models;
- recovery maps in quantum approximate recovery;
- gluing/global sections in sheaf contextuality;
- action/coupling rules in variational dynamics.

Definition 9.3 — Bridge-profile match

An underdeclared transfer $X \Rightarrow Y$ has a **bridge-profile match** with a solved bridge $\mathfrak{B}^*$ if there are structure-type maps:

$$\Phi_S : S_X \rightarrow S_\star$$

and:

$$\Phi_T : T_Y \rightarrow T_\star$$

such that the target distinction types and adequacy conditions correspond.

This does not establish the bridge. It identifies a candidate template.

Proposition 9.4 — Bridge-template transfer is candidate, not proof

If $\mathfrak{B}^*$ is a solved bridge template with a matching bridge profile for $X \Rightarrow Y$, then $\mathfrak{B}^*$ supplies candidate bridge data for $X \Rightarrow Y$. It does not establish adequacy unless the translated bridge satisfies:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}}(D_X).$$

Proof

A profile match identifies structural similarity between bridge tasks. But adequacy is defined by recovery of target distinctions in the target case. Unless that inclusion is verified, the transfer remains a candidate. $\square$

10. Calibration of bridge completions

Bridge completion is a learning process.

Definition 10.1 — Bridge completion attempt

A **bridge completion attempt** is a candidate bridge declaration $\mathfrak{B}_{X \rightarrow Y}$ proposed for an underdeclared transfer.

Definition 10.2 — Bridge calibration

A bridge completion attempt is **calibrated** against a target distinction family D'_Y by checking:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}}(D_X).$$

If the inclusion holds, the bridge succeeds for D'_Y . If it fails, the bridge failure is informative: it identifies distinctions not preserved by the candidate bridge.

Proposition 10.3 — Failed bridge calibration is valid learning

If a bridge completion attempt fails for D'_Y , then the failure is not meaningless. It identifies a missing preservation component or a mismatch between the solved bridge template and the underdeclared transfer.

Proof

Failure of bridge adequacy means there exists:

$$(y, y') \in D'_Y$$

such that:

$$(y, y') \notin \text{Rec}_{\mathfrak{B}}(D_X).$$

This identifies a target distinction not preserved by the attempted bridge. The failure therefore refines the bridge profile or shows that the template is inadequate. $\square$

11. No-go network generated by No Undeclared Relation

This section makes explicit a structure that is implicit across the companion programme.

The principle

No declared relation, no legitimate relational inference

does not stand alone. It generates a network of no-go families at different closure-depths. These depths are not a linear hierarchy. They form a preorder of dependence, realization, and bridgeability. Some no-gos sit on shared plains; some are incomparable; some are projections of the same primitive failure through different charted contexts.

The purpose of this section is not to claim that the programme's machinery is complete. It is not. Some relations, bridge types, and closure-depths remain unexcavated. The purpose is to show that the no-gos already produced by the companion papers are not scattered observations. They are instances of one relation-first discipline.

11.1 Closure-depth warning

A no-go network is not a numbered ladder:

$$L_0 < L_1 < L_2 < \dots.$$

It is better represented by a closure-depth preorder:

$$X \preceq_{\text{cl}} Y,$$

meaning that Y is obtainable from X by declared operations, traces, charting, bridge declarations, residual readouts, compression, or closure. Two structures may be closure-equivalent, incomparable, or mutually reachable through different traces.

Therefore the levels below are organizational strata, not ontological ranks. A theorem appearing at a later stratum is not necessarily more real, more fundamental, or less primitive. It may simply require more declared relation structure.

11.2 Relation-work checklist

Any asserted arrow

$$A \rightarrow B$$

is a relation claim. In this programme, such a claim is legitimate only when the relation declares at least:

1. source context A ;
2. target context B ;
3. source distinctions D_A ;
4. target distinctions D_B ;
5. preservation or recovery map;
6. admissibility condition;
7. failure mode;
8. boundary/definitional exceptions, if any.

If any of these are absent, the arrow is underdeclared. It may still be a useful heuristic, but it is not yet a valid relation-first inference.

11.3 No total preservation under proper transfer

The following theorem extracts a recurring pattern from Paper B, Charted Organization, Recursive Chart Learning, Cross-Closure, and the present note.

Definition 11.3.1 — Proper transfer

Let $\mathfrak{B}_{A \rightarrow B}$ be a bridge relation from closure context A to closure context B . It is a **proper transfer** relative to D_A if it is not a D_A -equivalence; that is, at least one source distinction in D_A is not recoverable in B under the bridge.

Equivalently:

$$D_A \not\subseteq \text{Rec}_{\mathfrak{B}^{-1}}(D_B),$$

when the reverse recovery expression is defined.

A quotient, compression, coarse chart, lossy bridge, or task-relative reduction is a proper transfer when it collapses at least one source distinction.

Definition 11.3.2 — Total preservation

A bridge $\mathfrak{B}_{A \rightarrow B}$ is **totally preserving relative to D_A** if every source distinction is recoverable on the target side:

$$D_A \subseteq \text{Rec}_{\mathfrak{B}^{-1}}(D_B).$$

Theorem 11.3 — No total preservation under proper transfer

A bridge relation cannot be both a proper transfer relative to D_A and totally preserving relative to D_A .

Proof

A proper transfer relative to D_A means at least one source distinction in D_A is not recoverable on the target side. Total preservation relative to D_A means every source distinction in D_A is recoverable on the target side. These conditions are contradictory. $\square$

Remark 11.3.3 — Relation to Paper B quotient admissibility

Theorem 11.3 is the cross-context version of Paper B's quotient discipline. Paper B states that a quotient is admissible only when it collapses no distinction visible to the declared task and structure type:

$$\sim_q \subseteq \sim_{\mathcal{T}}^R.$$

Equivalently, a quotient may be lossy only relative to distinctions the task has declared irrelevant. Theorem 11.3 generalizes the same discipline to bridge relations: a proper transfer is allowed to be lossy, but it cannot simultaneously be described as totally preserving the source distinctions. If a claimed transfer preserves all source distinctions, it should be classified as equivalence, enrichment, embedding, or wrong-direction relation rather than as a proper lossy transfer.

Corollary 11.4 — Direction diagnostic

If a claimed relation $A \rightarrow B$ preserves every A -distinction and B contains additional distinctions not recoverable from A , then the relation is not a proper compression of A into B . It is an embedding, enrichment, equivalence-with-extra-structure, or a relation whose claimed direction requires additional justification.

In particular, a theory claiming that A produces B must show whether:

1. B is a lossy/proper transfer of A ;
2. B is an enriched extension of A ;
3. A and B are equivalent relative to the declared task;
4. the relation actually runs in the opposite or a bidirectional direction;
5. extra bridge data supply the additional B -distinctions.

This corollary is one of the strongest practical uses of No Undeclared Relation: the direction of a relation is itself a declared structural claim, not a free arrow.

11.5 Network family I — Boundary no-gos

Boundary no-gos are the earliest relation-first restrictions.

No-go family	Representative form	Repair data
no boundaryless organization	organization cannot live at pure z_- or z_+	proper zero-boundary interior
no terminal completion as final object	$z_+ \equiv_{\partial} z_-$ at boundary collapse	recurrence/closure mediation
no proper organization at excluded extremes	$\rho(a, q) \in \{z_-, z_+\}$ is not proper	interior threshold / admissible cut

These no-gos originate in the Theory of Organization and Triadic Closure. Their common form is:

no stable relation without boundary discipline.

11.6 Network family II — Quotient and preservation no-gos

These are Paper B / task-visible quotient no-gos.

No-go family	Representative form	Repair data
no quotient admissibility without task refinement	$\sim_q \not\subseteq \sim_{\mathcal{T}}$	declared task-visible equivalence

No-go family	Representative form	Repair data
no preservation without loss/readout declaration	target distinctions not tied to loss	faithful loss / recovery map
no free informativeness comparison	representation A vs B not comparable by slogan	resource preorder / Blackwell-like comparison

Their common form is:

no preservation claim without declared preserved distinctions.

11.7 Network family III — Triadic closure no-gos

These are the instability no-gos from Triadic Closure.

Missing relation-mode	Failure
boundary/polarity missing	no extremal discipline
mediation/path missing	no resolving relation
visibility/evaluation missing	no chart-visible result

The shared theorem is:

boundary + mediation + visibility

is the first closure-complete arity relative to the programme's declared closure requirements. Proper subfamilies are closure-incomplete. Higher arities may refine the schema, but they do not eliminate the need for these relation-modes unless additional primitive requirements are introduced.

11.8 Network family IV — Chart and observer no-gos

These are Charted Organization no-gos.

No-go family	Representative form	Repair data
no chartless information	distinction not chart-visible is not information	admissible observer-chart
no observer from outside	observer must be relation-internal	positioned sub-chart
no arbitrary chart imposition	imposed readout must be admissible	relation-map / chartability data

No-go family	Representative form	Repair data
no single-chart objectivity	identity transition gives vacuous invariance	nontrivial transition class

Common form:

no visibility without an admissible chart.

11.9 Network family V — Objecthood and gluing no-gos

These are Charted Objecthood / contextual gluing no-gos.

No-go family	Representative form	Repair data
no local-to-global objecthood for free	local proper organizations do not imply global object	compatible family / global section
no objecthood under fixed-point-free holonomy	$H_\gamma(O) \neq O$ for all O	flat or fixed-point-compatible transport
no single-chart objecthood	one chart gives fixation, not objecthood	nonvacuous atlas compatibility

Common form:

no global object without gluing relation.

11.10 Network family VI — Learning and trace no-gos

These are Recursive Chart Learning no-gos.

No-go family	Representative form	Repair data
no learning without admitted landed trace	dangling attempt is not a trace	landed admissible transition
no oracle evaluation	evaluator cannot use undeclared unavailable outcomes	chart-visible branch field / declared task data
no endpoint equality as learning equality	same endpoint does not recover trace	path trace
no residual learning without faithful readout	residual alone is not learning pressure	loss/evaluator/admission bridge

No-go family	Representative form	Repair data
no stability past support margin	perturbation beyond margin exits guarantee	recalibration / new threshold policy

Common form:

no learning without declared trace, evaluation, and admissibility.

11.11 Network family VII — Residual and dynamics no-gos

These no-gos bridge Triadic Closure and Recursive Chart Learning.

No-go family	Representative form	Repair data
no residual activation without readout	Ω alone does not imply update	faithful residual readout
no residual-to-force without coupling	curvature/defect is not dynamics	action/coupling rule
no residual-to-learning without evaluator	mismatch is not pressure	evaluator/admission region
no gauge group from residual-preservation alone	symmetry-like preservation is not physical gauge theory	composition, identity, inverse, action, covariance, representation

Common form:

no active consequence from residual without a declared coupling/readout relation.

11.12 Network family VIII — External bridge no-gos

These are the external-facing no-gos of the present paper.

Transfer	No-go	Required bridge
entropy $\rightarrow$ geometry	entropy values do not freely transfer to metric/causal structure	dictionary, area law, region class, code subspace, dynamics
recoverability $\rightarrow$ causality	dependence is not intervention	causal model, counterfactual/intervention semantics

Transfer	No-go	Required bridge
positive metric $\rightarrow$ Lorentzian	positive inertia cannot become indefinite by real reparametrization	causal/order/indefinite/Wick/Krein bridge
prediction $\rightarrow$ explanation	predictive sufficiency is not explanatory sufficiency	explanation task, causal bridge, interpretability loss
local compatibility $\rightarrow$ objecthood	local satisfiability is not global gluing	sheaf/transport/global-section bridge

Common form:

no external transfer without explicit bridge relation.

11.13 The generated no-go network

The no-go families above are linked by one relation-first generator:

No declared relation, no legitimate relational inference.

Different companion papers instantiate this generator at different closure-depths:

Context	No-go expression
boundary	no organization at excluded extrema
quotient	no task preservation under improper quotient
triad	no closure-complete dyad
chart	no information without chart visibility
objecthood	no global object without gluing
learning	no learning without admitted trace
residual	no dynamics without readout/coupling
external transfer	no target distinction without bridge

Thus the programme's no-gos are not isolated. They are projections of the same primitive relation discipline into different charted contexts.

11.14 Repair map

Every no-go has a corresponding repair form.

No-go	Repair
undeclared relation	declare source, target, preservation, admissibility
improper quotient	declare task-visible equivalence and show refinement
boundary collapse	move to proper interior
dyadic instability	supply missing boundary, mediation, or visibility role
chartless distinction	supply admissible observer-chart
local-only object	supply gluing/transport compatibility
dangling attempt	supply landed admissible transition
residual inactivity	supply faithful readout/coupling/evaluator
positive-to-Lorentzian failure	supply order/indefinite/signature bridge

This is the constructive side of the no-go network. The point of the no-go is not to forbid theory-building; it is to specify what must be declared for theory-building to become inspectable.

11.15 Excavation status

This network covers the programme’s currently articulated no-go content at the level needed for the present paper.

It is not claimed to be final or exhaustive. The programme may contain unexcavated no-gos, unclassified bridge types, and future refinements of the families listed above. Moreover, the closure-depth structure is not a linear sequence; several no-gos occupy shared plains or can be reached by different traces. The table above is therefore a current relation-first map, not a final atlas.

The methodological claim is:

the no-gos currently produced by the programme form a coherent relation-first network generated by No Undecl

Future work may refine the network, add missing bridge types, or show that some listed no-gos are instances of still more primitive constraints.

12. Bridge-type taxonomy

The following bridge types recur across foundations claims.

Bridge type	Preserves	Mature solved / declared examples
chartability bridge	local relation data to charted organization	chart face maps and functional resolution
faithful readout bridge	residual to loss/evaluator	proper scoring rules, faithful loss modes
causal bridge	dependence to causal precedence	structural causal models, intervention calculus
order-volume bridge	order to Lorentzian/conformal/scale structure	causal set theory, causal reconstruction
dictionary bridge	boundary data to bulk data	AdS/CFT dictionary, holographic reconstruction
gluing bridge	local compatibility to global object	sheaf/global-section structures
recovery bridge	partial data to full state/structure	Petz/Fawzi-Renner recovery, sufficient statistics
variational/coupling bridge	residual to dynamics	action principles, control update rules
threshold bridge	trace thresholds to persistence supports	persistence-compatible trace schedules
comparison bridge	one value space to another	declared comparison maps or bridge charts

13. Representative diagnostic table

The following table applies the no-free-transfer diagnostic to common foundations moves.

Claimed transfer	Source context	Target context	Required bridge data
Entropy $\Rightarrow$ geometry	entropy / recoverability	metric or causal geometry	holographic dictionary, area law, region class, code subspace, reconstruction theorem, dynamics
Recoverability $\Rightarrow$ causality	inference / prediction	causal precedence	intervention model, causal Markov assumptions, temporal/order assumptions

Claimed transfer	Source context	Target context	Required bridge data
Positive information metric $\Rightarrow$ Lorentzian metric	distinguishability geometry	indefinite causal metric	causal order, hyperbolic dynamics, Wick/Krein structure, order-volume data
Prediction $\Rightarrow$ explanation	predictive sufficiency	interpretability or causality	explanation task, causal bridge, stable attribution criterion
Local compatibility $\Rightarrow$ global object	local chart agreement	objecthood	gluing/transport/transition compatibility
Computation $\Rightarrow$ physical law	computational transition	physical dynamics	physical realization, conservation constraints, energetic/covariance data
Gauge redundancy $\Rightarrow$ physical equivalence	representational symmetry	physical sameness	action, observables, covariance, quotient discipline
Residual $\Rightarrow$ dynamics	closure defect	update/force/learning pressure	faithful residual readout, evaluator, admission region
Learning gain $\Rightarrow$ truth	evaluator improvement	correctness or ontology	faithful loss, calibrated task, adequacy bridge

The table does not refute these transfers. It states what must be declared before each transfer becomes inspectable.

14. Case study I — Entropy to geometry

14.1 Transfer claim

entropy / entanglement data $\Rightarrow$ geometry.

14.2 Audit chart

$Q_1 : X =$ entropy or recoverability context.

$Q_2 : Y =$ geometric context.

$Q_3 : D_Y = \text{metric, causal, area, volume, topological, or dynamical distinctions.}$

$Q_4 : \text{Entropy equality alone does not encode all geometric distinctions.}$

$Q_5 : \text{required bridge data include dictionary, region class, area law, code subspace, causal/dynamical assumptions}$

14.3 Solved bridge templates

A mature bridge template appears in holographic settings: the Ryu–Takayanagi formula relates boundary entanglement entropy to bulk minimal surface area in AdS/CFT; the AdS/CFT dictionary relates boundary CFT data to bulk gravitational data. These are bridge declarations, not consequences of entropy alone.

Another bridge template appears in causal set theory: causal order supplies conformal/causal structure while cardinality supplies volume information in continuum approximation.

14.4 Candidate completion insight

An entropy-to-geometry programme should not ask only whether entropy values are preserved. It should ask which extra bridge supplies:

1. order/cone information;
2. scale/volume information;
3. metric reconstruction;
4. dynamics.

The causal-set template suggests a diagnostic:

geometry requires at least order-like and scale-like bridge data.

Entropy may contribute to one or both, but the bridge must declare how.

14.5 Claim

The constructive conclusion is not that entropy-to-geometry programmes fail. It is that compressed claims of the form:

$$\text{entropy} \Rightarrow \text{geometry}$$

must be expanded into their bridge declarations before they are inspectable. In careful holographic work, much of this bridge data is explicitly supplied: dictionary, region class, code subspace, area

relation, reconstruction theorem, and dynamical setting. The audit is therefore a vulnerability check on compressed slogans, not a critique of responsible technical literature.

an entropy-to-geometry derivation is inspectable exactly to the extent that its geometric bridge data are declared

15. Case study II — Positive information geometry to Lorentzian structure

15.1 Transfer claim

positive distinguishability geometry $\Rightarrow$ Lorentzian signature.

15.2 Audit chart

$Q_1 : X = \text{positive information/recoverability geometry.}$

$Q_2 : Y = \text{Lorentzian metric structure.}$

$Q_3 : D_Y = \text{indefinite signature, causal cones, time-like direction, conformal factor.}$

$Q_4 : \text{positive quadratic forms do not encode negative directions under real nondegenerate reparametrization.}$

$Q_5 : \text{missing bridge: causal order, hyperbolic dynamics, Wick/Krein structure, order-volume data, or indefinite p}$

15.3 Solved bridge templates

- Causal set theory declares causal order and local finiteness; in continuum approximation, order corresponds to causal structure and cardinality to volume.
- Wick rotation declares analytic continuation.
- Lorentzian noncommutative geometry declares Krein, temporal, twist, or causal-cone structures.
- Hyperbolic dynamics declares a propagation/causal structure.

15.4 Candidate completion insight

A positive-information programme needs a declared order/indefinite bridge. The completion cannot be “more information geometry” if the target distinction is Lorentzian signature.

The bridge audit yields:

the Lorentzian bridge must contain the first non-positive or order-like ingredient.

16. Case study III — Recoverability to causality

16.1 Transfer claim

recoverable dependence $\Rightarrow$ causal precedence.

16.2 Audit chart

$Q_1 : X =$ recoverability, prediction, correlation, reconstruction.

$Q_2 : Y =$ causal precedence or intervention structure.

$Q_3 : D_Y =$ direction of influence, counterfactual response, intervention effect.

$Q_4 :$ recoverability does not by itself encode intervention direction.

$Q_5 :$ missing bridge: intervention semantics, structural causal model, temporal/order assumption.

16.3 Solved bridge template

Structural causal models declare intervention structure and causal equations. They do not infer causality from prediction alone; they add a bridge that distinguishes observation from intervention.

16.4 Candidate completion insight

Any programme moving from recoverability to causality needs an intervention or counterfactual bridge. Otherwise it remains within recoverability.

17. Case study IV — Local compatibility to global objecthood

17.1 Transfer claim

local compatibility $\Rightarrow$ global object.

17.2 Audit chart

$Q_1 : X = \text{local chart organization.}$

$Q_2 : Y = \text{global objecthood.}$

$Q_3 : D_Y = \text{compatible global family across all relevant charts.}$

$Q_4 : \text{local consistency does not encode global gluing.}$

$Q_5 : \text{missing bridge: transport, gluing, global-section compatibility.}$

17.3 Solved bridge template

Sheaf-theoretic contextuality gives a mature gluing obstruction: local sections may exist on all contexts while no global section exists. The charted objecthood framework generalizes this as objecthood obstruction across observer-chart atlases.

17.4 Candidate completion insight

The solved bridge suggests that local compatibility claims should be audited for:

1. local sections;
2. restriction or transport maps;
3. compatibility on overlaps or transitions;
4. global section existence.

Without those, local compatibility is not global objecthood.

18. Case study V — Residual to dynamics

18.1 Transfer claim

$\text{residual} \Rightarrow \text{dynamics or learning pressure.}$

18.2 Audit chart

$Q_1 : X = \text{closure residual, holonomy, anomaly, mismatch, loss.}$

$Q_2 : Y = \text{update, dynamics, force, learning pressure.}$

$Q_3 : D_Y =$ direction of update, admissible transition, dynamical effect.

$Q_4 :$ residual magnitude alone does not encode an update rule.

$Q_5 :$ missing bridge: faithful readout, evaluator, admission region, variational/coupling rule.

18.3 Solved bridge templates

- Variational mechanics turns action variation into equations of motion by declaring a variational principle.
- Gauge theory turns curvature into dynamics only once an action/coupling and field equations are declared.
- Recursive Chart Learning turns residual into learning pressure only through faithful loss readout, evaluator, and admissible region.
- Control theory turns error into update through a controller or feedback law.

18.4 Candidate completion insight

Residuals become active only through a declared coupling/readout/update bridge. A nonzero residual is not yet dynamics.

19. Case study VI — Prediction to explanation

19.1 Transfer claim

predictive success $\Rightarrow$ explanation or understanding.

19.2 Audit chart

$Q_1 : X =$ predictive representation.

$Q_2 : Y =$ explanatory, causal, or interpretable representation.

$Q_3 : D_Y =$ counterfactual stability, feature meaning, causal mechanism, human-usable decomposition.

$Q_4 :$ prediction does not automatically encode these distinctions.

Q_5 : missing bridge: interpretability task, causal task, explanation criterion, faithful explanation loss.

19.3 Solved bridge templates

- Causal inference supplies intervention/counterfactual semantics.
- Interpretability methods supply explanation tasks and stability criteria.
- Information bottleneck supplies a relevance variable for compression.
- Rate-distortion supplies declared distortion.

19.4 Candidate completion insight

Prediction-to-explanation requires declaring which explanatory distinctions matter. A predictive quotient may be minimal for prediction and insufficient for explanation.

20. Comparative bridge-completion table

Underdeclared case	Missing bridge type	Solved template	Candidate insight
Entropy $\rightarrow$ geometry	dictionary/order/scale bridge	AdS/CFT, causal sets	geometry needs order-like and scale-like bridge data
Positive information metric $\rightarrow$ Lorentzian	indefinite/order bridge	causal order, Wick/Krein, Lorentzian NCG	positivity must be supplemented by causal/signature structure
Recoverability $\rightarrow$ causality	intervention bridge	structural causal models	prediction is not intervention
Local compatibility $\rightarrow$ objecthood	gluing bridge	sheaf/global sections	local satisfiability is not global object
Residual $\rightarrow$ dynamics	coupling/update bridge	variational mechanics, control, learning loss	residual must be read as actionable
Prediction $\rightarrow$ explanation	explanation/causal bridge	interpretability, causal models, IB/RD	predictive sufficiency is not explanatory sufficiency

21. Result-type distribution

Using the success scale:

modest = classifies the missing bridge,

stronger = finds a solved bridge template with matching bridge profile,

strongest = uses the solved bridge to produce a new theorem, diagnostic, or constraint in the underdeclared case.

The present note contains:

Case study	Current level
Entropy $\rightarrow$ geometry	stronger
Positive information geometry $\rightarrow$ Lorentzian	stronger, with existing no-go support
Recoverability $\rightarrow$ causality	stronger
Local compatibility $\rightarrow$ objecthood	strongest inside the programme
Residual $\rightarrow$ dynamics / learning pressure	strongest inside the programme; stronger externally
Prediction $\rightarrow$ explanation	modest-to-stronger

The paper therefore mostly establishes a constructive methodology and identifies solved bridge templates. Its strongest results are where the companion programme already supplies theorem-level bridge completions.

22. Worked bridge-completion attempts

This section closes the bridge-audit loop for four representative transfers. Each attempt follows the same pattern:

audit $\rightarrow$ candidate bridge $\rightarrow$ calibration $\rightarrow$ success or failure diagnosis.

The purpose is not to prove that all external transfers succeed. The purpose is to show how the No-Free-Transfer theorem becomes constructive: when the bridge succeeds, the target distinctions are recovered; when the bridge fails, the failure identifies the missing bridge data.

Each worked attempt should be read as a source–bridge–target triad:

$X - \mathfrak{B}_{X \rightarrow Y} - Y.$

The examples are included not merely to name missing bridges, but to show that the bridge is the necessary mediating relation. Source and target alone do not specify what is preserved, compared, recovered, or erased.

22.1 Attempt I — Residual to learning pressure

Triadic form.

$$\boxed{\Omega - \mathfrak{B}_{\Omega \rightarrow L} - L_\tau.}$$

The residual context is the source, learning pressure is the target, and the faithful residual-to-loss/evaluator/admission bridge is the mediating relation.

Source context

Let:

$$X = \Omega$$

be a residual context. Its source distinctions are residual distinctions:

$$D_X = \{(\Omega_1, \Omega_2) : \Omega_1 \not\equiv_\Omega \Omega_2\}.$$

Target context

Let:

$$Y = L_\tau$$

be a learning-pressure context for a task τ . Its target distinctions include:

$$D_Y = \{\text{admitted vs. non-admitted, improving vs. regressive, traceable vs. non-traceable}\}.$$

Candidate bridge

Declare a residual-to-learning bridge:

$$\mathfrak{B}_{\Omega \rightarrow L} = (\text{Loss}_\tau, E_\tau, W_\tau^{\text{adm}}, X_\tau),$$

where:

- Loss_τ is a faithful readout of residual structure;
- E_τ is an evaluator induced by the loss, for example by order reversal;
- W_τ^{adm} is the admitted evaluator region;
- X_τ is the declared branch or trace domain.

Calibration test

The bridge is adequate for a target distinction family $D'_Y \subseteq D_Y$ only if:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}_{\Omega \rightarrow L}}(D_X).$$

In concrete terms, if two residuals satisfy:

$$\Omega_1 \not\equiv_\Omega \Omega_2,$$

then the bridge transfers that distinction to learning pressure only when:

$$\text{Loss}_\tau(\Omega_1) \neq \text{Loss}_\tau(\Omega_2)$$

and that difference is visible to E_τ and relevant to W_τ^{adm} or the outcome classes.

Success theorem

Theorem 22.1 — Faithful residual readout induces learning-pressure distinction Suppose:

1. $(\Omega_1, \Omega_2) \in D_X$;
2. Loss_τ is faithful for the residual distinction, so

$$\Omega_1 \not\equiv_\Omega \Omega_2 \Rightarrow \text{Loss}_\tau(\Omega_1) \neq \text{Loss}_\tau(\Omega_2);$$

3. E_τ is induced by Loss_τ and preserves the relevant ordering or class separation;
4. the induced evaluator values lie in declared admission/outcome regions.

Then the residual distinction transfers to a learning-pressure distinction.

Proof

By faithfulness, the residual distinction is visible as a loss distinction. Since E_τ is induced by the loss and preserves the declared ordering or class separation, the loss distinction is visible as an evaluator distinction. If the evaluator distinction intersects the admission or outcome classes, it determines a learning-pressure distinction. Hence the target learning-pressure distinction lies in:

$$\text{Rec}_{\mathfrak{B}_{\Omega \rightarrow L}}(D_X).$$

□

Failure theorem

Theorem 22.2 — Non-faithful residual readout blocks learning-pressure transfer If:

$$\Omega_1 \not\equiv_{\Omega} \Omega_2$$

but:

$$\text{Loss}_{\tau}(\Omega_1) = \text{Loss}_{\tau}(\Omega_2),$$

then this residual distinction does not transfer to learning pressure through Loss_{τ} .

Proof

The candidate bridge collapses the residual distinction at the readout stage. Since the evaluator is induced from the loss, no evaluator distinction can recover a distinction already erased by the loss unless additional bridge data are supplied. Therefore:

$$(\Omega_1, \Omega_2) \notin \text{Rec}_{\mathfrak{B}_{\Omega \rightarrow L}}(D_X)$$

for the claimed learning-pressure distinction. □

Diagnosis

The failed bridge identifies the missing component:

a faithful residual-to-loss readout is missing.

This is a theory-native learning result. A nonzero residual is not yet learning pressure. It becomes learning pressure only under a faithful readout/evaluator/admission bridge.

22.2 Attempt II — Local compatibility to global objecthood

Triadic form.

$$\{\text{Org}_{\theta_e}^{\circ}(\mathcal{C})\}_{\mathcal{C}} - \mathfrak{B}_{\text{glue}} - \text{Obj}(\mathfrak{A}_R).$$

Local proper organization is the source, global objecthood is the target, and gluing/transport compatibility is the mediating relation.

Source context

Let:

$$X = \{\text{Org}_{\theta_e}^\circ(\mathcal{C})\}_{\mathcal{C} \in \mathbf{C}_R}$$

be the family of local proper organization layers over an observer atlas.

Target context

Let:

$$Y = \text{Obj}(\mathfrak{A}_R)$$

be global objecthood: compatible charted organization across the atlas.

Target distinctions include:

$$D_Y = \{\text{compatible family vs. no compatible family, objecthood vs. local fixation}\}.$$

Candidate bridge

Declare a gluing bridge:

$$\mathfrak{B}_{\text{glue}} = (\text{Ctx}_{\mathfrak{A}}, \mathcal{F}_{\text{Org}}, \rho_{\mathcal{CD}}),$$

where:

- $\text{Ctx}_{\mathfrak{A}}$ is the chart context category;
- $\mathcal{F}_{\text{Org}}$ assigns each chart its proper organization layer;
- $\rho_{\mathcal{CD}}$ are restriction or transport maps;
- compatibility means agreement under all declared transition/restriction maps.

Success theorem

Theorem 22.3 — Gluing bridge turns local compatibility into objecthood If a family:

$$(O_{\mathcal{C}})_{\mathcal{C} \in \mathbf{C}_R}$$

satisfies:

$$\rho_{\mathcal{CD}}(O_{\mathcal{D}}) = O_{\mathcal{C}}$$

for every declared restriction or transport relation, then the family defines a charted object.

Proof

The stated condition is exactly the compatible-family condition for the organization presheaf or transport functor. By the charted objecthood definition, compatible families of local proper organizations are charted objects. $\square$

Failure theorem

Theorem 22.4 — Local proper organization does not transfer to objecthood without gluing If every chart $\mathcal{C}$ has at least one local proper organization but no compatible family exists across declared transitions, then local compatibility does not transfer to global objecthood.

Proof

Objecthood requires a compatible family. Local nonemptiness gives:

$$\text{Org}_{\theta_e}^\circ(\mathcal{C}) \neq \emptyset$$

for every chart, but this does not imply that a family can be chosen satisfying all transition constraints. If no such family exists, the target objecthood distinction is not recovered. The No-Free-Transfer theorem applies. $\square$

Holonomy diagnosis

If a cycle γ in the chart context induces holonomy:

$$H_\gamma : \text{Org}_{\theta_e}^\circ(\mathcal{C}) \rightarrow \text{Org}_{\theta_e}^\circ(\mathcal{C})$$

and:

$$H_\gamma(O) \neq O$$

for every local candidate O , then no compatible object family can have base value O .

Thus the missing bridge component is:

flat or fixed-point-compatible gluing transport.

22.3 Attempt III — Positive information geometry to Lorentzian structure

Triadic form.

$G_+ - \mathfrak{B}_{+\rightarrow L} - G_L.$

Positive information geometry is the source, Lorentzian structure is the target, and the required order/indefinite/analytic/signature bridge is the mediating relation.

Source context

Let:

$$X = G_+$$

be a positive or positive-semidefinite information-geometric context, such as one generated by Fisher-type, Bures-type, relative-entropy-Hessian, or positive recoverability geometry.

Its source distinctions are distinguishability or recovery-loss distinctions.

Target context

Let:

$$Y = G_L$$

be Lorentzian metric structure. Its target distinctions include:

$$D_Y = \{\text{indefinite signature, time-like direction, causal cones, conformal causal order, metric scale}\}.$$

No-free-transfer diagnosis

Positive distinguishability data do not encode negative directions under real nondegenerate reparametrization. Therefore:

$$G_+ \not\Rightarrow G_L$$

without additional bridge data.

Candidate bridge types

Possible solved bridge templates include:

1. **causal/order bridge**: causal order or precedence structure;
2. **order-volume bridge**: causal order plus volume/cardinality data;
3. **Wick/analytic bridge**: analytic continuation or signature-changing operation;
4. **Krein/indefinite bridge**: indefinite pairing or Lorentzian spectral structure;
5. **hyperbolic-dynamical bridge**: propagation equations or domain-of-dependence structure.

Failure theorem

Theorem 22.5 — Positive information geometry alone does not recover Lorentzian target distinctions Let G_+ supply only positive or positive-semidefinite quadratic-form distinctions. Let $D'_Y \subseteq D_Y$ include indefinite signature. If no bridge supplies causal, indefinite, analytic, Wick, Krein, order-volume, or hyperbolic structure, then:

$$D'_Y \not\subseteq \text{Rec}_{\mathfrak{B}}(D_X).$$

Therefore $G_+ \not\Rightarrow_{D'_Y} G_L$.

Proof

A positive-semidefinite quadratic form has no negative direction. Under real nondegenerate reparametrization, the inertia of a symmetric bilinear form is preserved. Therefore an indefinite Lorentzian signature is not recoverable from positive quadratic-form data alone. Since no bridge supplies an indefinite/order-like/signature-changing ingredient, the target distinction is not recovered. The No-Free-Transfer theorem applies. $\square$

Conditional success theorem

Theorem 22.6 — An order/indefinite bridge can make Lorentzian transfer inspectable If a bridge declaration $\mathfrak{B}_{+\rightarrow L}$ supplies an order-like or indefinite structure sufficient to recover the target Lorentzian distinction family D'_Y , then the transfer:

$$G_+ + \mathfrak{B}_{+\rightarrow L} \Rightarrow G_L$$

is inspectable by checking:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}_{+\rightarrow L}}(D_X).$$

Proof

This is bridge adequacy. Once the bridge supplies the target distinctions and the inclusion is verified, the cross-context inference is no longer free transfer. $\square$

Diagnosis

The failure is not “information geometry is useless.” The failure is:

positive distinguishability does not itself provide Lorentzian signature.

The next step is to identify and declare the first causal, indefinite, analytic, or order-volume ingredient.

22.4 Attempt IV — Featured quantitative theorem: persistence margin to trace stability

Triadic form.

$$\boxed{P - \mathfrak{B}_{P \rightarrow T} - T_\tau.}$$

Persistence support is the source, path-trace stability is the target, and the persistence-margin/threshold-policy/transport bridge is the mediating relation.

This attempt is the quantitative bridge-completion example and the strongest theorem-grade success case in this paper. It is the quantitative version of the persistence-compatible trace bridge from Cross-Closure Instantiation Theorems: persistence support comes from the Theory of Organization, charted organization from Chartist Organization Theory, path traces and threshold policies from Recursive Chart Learning, and residual perturbations from Triadic Closure / Cross-Closure.

Source context

Let:

$$X = P$$

be a persistence-support context. A local organization O_i in chart $\mathcal{C}_i$ has a persistence support interval:

$$I_i = (\ell_i, u_i]$$

inside a metric zero-boundary value space $Z_{\mathcal{C}_i}$.

The source distinctions are persistence distinctions:

$$D_X = \{\theta \in I_i \text{ vs. } \theta \notin I_i\}.$$

Target context

Let:

$$Y = T_\tau$$

be a learning path-trace context. The target distinction is:

trace remains proper vs. trace exits the proper organization layer.

A path trace is:

$$p : \mathcal{C}_0 \xrightarrow{\phi_0} \mathcal{C}_1 \xrightarrow{\phi_1} \dots \xrightarrow{\phi_{n-1}} \mathcal{C}_n.$$

A threshold policy supplies:

$$\Theta_\tau(p) = (\theta_0, \dots, \theta_n).$$

At each chart $\mathcal{C}_i$, the transported organization is proper when:

$$\theta_i \in I_i.$$

Candidate bridge

Declare a persistence-to-trace bridge:

$$\mathfrak{B}_{P \rightarrow T} = (I_i, \Theta_\tau, p, \delta_i, \text{Trans}_i),$$

where:

- $I_i = (\ell_i, u_i]$ is the persistence support of the transported organization at $\mathcal{C}_i$;
- $\Theta_\tau(p)$ is the threshold schedule along the trace;
- δ_i is a residual perturbation or threshold displacement at step i ;
- Trans_i is the declared transport identifying the organization at adjacent charts.

Define the persistence margin:

$$m_i = \text{dist}(\theta_i, \partial I_i) = \min(\theta_i - \ell_i, u_i - \theta_i)$$

when $Z_{\mathcal{C}_i}$ is realized as an interval in $\mathbb{R}$.

For half-open supports $(\ell_i, u_i]$, the lower boundary is excluded and the upper boundary is included. Therefore exact boundary cases require the convention:

$$\theta_i - \delta_i > \ell_i, \quad \theta_i + \delta_i \leq u_i.$$

The strict version

$$|\delta_i| < m_i$$

is sufficient and avoids endpoint ambiguity.

Success theorem

Theorem 22.7 — Persistence margin stabilizes proper learning traces Let p be a path trace with transported organizations O_i and threshold schedule:

$$\Theta_\tau(p) = (\theta_0, \dots, \theta_n).$$

Assume:

1. each O_i has persistence support $I_i = (\ell_i, u_i]$;
2. each threshold lies inside its support:

$$\theta_i \in I_i;$$

3. residual perturbations satisfy:

$$|\delta_i| < m_i = \min(\theta_i - \ell_i, u_i - \theta_i)$$

for every i ;

4. the transport maps identify the same path-traced organization along p .

Then the perturbed threshold schedule:

$$\Theta'_\tau(p) = (\theta_0 + \delta_0, \dots, \theta_n + \delta_n)$$

also remains inside the persistence supports:

$$\theta_i + \delta_i \in I_i$$

for every i . Therefore the path trace remains in the proper organization layer at every step.

Proof

For each i , the margin condition gives:

$$|\delta_i| < \theta_i - \ell_i$$

and:

$$|\delta_i| < u_i - \theta_i.$$

Thus:

$$\theta_i + \delta_i > \ell_i$$

and:

$$\theta_i + \delta_i < u_i$$

or, under the half-open endpoint convention, certainly:

$$\theta_i + \delta_i \leq u_i.$$

Therefore:

$$\theta_i + \delta_i \in (\ell_i, u_i] = I_i.$$

Since this holds for every i , every transported organization O_i remains proper at its perturbed threshold. Hence the whole trace remains inside the proper organization layer. $\square$

Failure theorem

Theorem 22.8 — Crossing the persistence margin blocks guaranteed trace stability If for some i :

$$|\delta_i| \geq m_i,$$

then persistence support alone no longer guarantees that:

$$\theta_i + \delta_i \in I_i.$$

Thus the persistence-to-trace bridge does not guarantee proper trace preservation at step i .

Proof

By definition:

$$m_i = \min(\theta_i - \ell_i, u_i - \theta_i).$$

If $|\delta_i| \geq m_i$, then there exists a perturbation direction that reaches or crosses one boundary of I_i . For negative perturbation it may reach ℓ_i , which is excluded. For positive perturbation it may exceed u_i . Therefore support membership is no longer guaranteed by the declared margin. $\square$

Recursive learning interpretation

This attempt is a concrete instance of Recursive Chart Learning.

A path trace is admitted under a threshold policy. If residual perturbations remain below the persistence margin, the trace stays viable and can update the future available step field. If the

perturbation crosses the margin, the trace may exit proper organization and the update loses its admissibility guarantee.

Thus the bridge produces a learning diagnosis:

$$\boxed{|\delta_i| < m_i \Rightarrow \text{trace viability is stable at step } i.}$$

$$\boxed{|\delta_i| \geq m_i \Rightarrow \text{next learning step requires recalibration, recharting, or a new threshold policy.}}$$

This is a quantitative example of the programme's learning loop:

path trace $\rightarrow$ evaluation by persistence margin $\rightarrow$ success or failure $\rightarrow$ updated bridge/threshold policy.

Worked numerical example

Let a transported organization have support:

$$I = (0.2, 0.9].$$

Let:

$$\theta = 0.5.$$

Then:

$$m = \min(0.5 - 0.2, 0.9 - 0.5) = \min(0.3, 0.4) = 0.3.$$

If:

$$|\delta| = 0.1,$$

then:

$$0.4 \in (0.2, 0.9], \quad 0.6 \in (0.2, 0.9],$$

so the trace remains proper.

If:

$$\delta = -0.31,$$

then:

$$\theta + \delta = 0.19 \notin (0.2, 0.9],$$

so the trace exits the persistence support. The failure identifies the missing next move: recalibrate the threshold policy, rechart the organization, or supply a different persistence bridge.

23. Branch traces for bridge completion

The worked attempts above show individual bridge calibrations. This section makes the branching structure explicit.

Every branch in a bridge-completion trace is a candidate middle term in a source–bridge–target triad. A branch succeeds when that middle term adequately recovers the target distinctions; it fails when that middle term is absent, ill-typed, too weak, non-faithful, or oriented in the wrong direction.

This is not a repetition of the No-Free-Transfer theorem. No-Free-Transfer states that a nontrivial target distinction cannot transfer without adequate bridge data. A **bridge-completion branch trace** records the actual sequence of candidate bridges tried, the success or failure of each branch, and the diagnostic repair indicated by each failure.

Thus the branch-trace layer answers a different question:

given a missing bridge, what candidate repairs are tried, and what does each result teach?

23.1 Bridge-completion branch

A **bridge-completion branch** for a transfer claim $X \Rightarrow Y$ is a tuple

$$b = (\mathfrak{B}_b, D'_Y, \text{Test}_b, \text{Out}_b),$$

where:

- $\mathfrak{B}_b$ is a candidate bridge declaration;
- $D'_Y \subseteq D_Y$ is the target distinction family being tested;
- Test_b is the declared adequacy test;
- Out_b is the branch outcome.

The branch is evaluated by checking:

$$D'_Y \subseteq \text{Rec}_{\mathfrak{B}_b}(D_X).$$

23.2 Branch outcomes

The outcome Out_b may be one of the following:

1. **success**: the candidate bridge recovers the target distinctions;
2. **partial success**: the candidate bridge recovers a proper subfamily of the target distinctions;
3. **failure**: the candidate bridge recovers none of the target distinctions or collapses the decisive distinction;
4. **wrong-direction relation**: the candidate bridge shows that the proposed arrow should be reversed, weakened, or reclassified;
5. **ill-typed comparison**: the source and target value spaces or distinction types are not comparable without additional comparison data;
6. **underdeclared branch**: the candidate bridge lacks enough data to run the adequacy test.

These outcomes are not psychological judgements. They are bridge-calibration statuses.

23.3 Bridge-completion trace

A **bridge-completion trace** is a finite rooted branch tree

$$\mathcal{T}_{X \rightarrow Y}$$

whose root is the transfer claim $X \Rightarrow Y$, whose nodes are bridge-completion branches, and whose edges are repair or refinement steps.

An edge

$$b_i \rightarrow b_j$$

means that b_j is a candidate repair, strengthening, weakening, or retyping of b_i based on the diagnostic output of b_i .

23.4 Diagnostic failure theorem

Theorem 23.1 — Failed branches are constructive when they isolate a missing bridge component

Let $b = (\mathfrak{B}_b, D'_Y, \text{Test}_b, \text{Out}_b)$ be a bridge-completion branch. Suppose b fails because a specific component c is absent from $\mathfrak{B}_b$, and suppose that every adequate bridge for D'_Y in the declared

bridge class must include a component of type c . Then the failure of b is constructive: any successful repair inside the declared bridge class must add, replace, or simulate c .

Proof

Because b fails, $D'_Y \not\subseteq \text{Rec}_{\mathfrak{B}_b}(D_X)$. By hypothesis, the failure is traced to the absence of component c , and every adequate bridge in the declared class requires a component of type c . Therefore any repaired bridge $\mathfrak{B}'$ that is adequate for D'_Y must contain, replace, or simulate c . Thus the failure identifies a necessary repair direction. $\square$

This theorem is distinct from No-Free-Transfer. No-Free-Transfer says that an inadequate bridge cannot establish the target. The present theorem says when the failure of a particular inadequate bridge is diagnostically constructive.

23.5 Branch trace I — Residual to learning pressure

Transfer claim

$$\Omega \Rightarrow L_\tau.$$

Target distinctions

$$D'_Y = \{\text{admitted vs. non-admitted, improving vs. regressive, traceable vs. non-traceable}\}.$$

Branch b_0 : residual alone

Candidate bridge:

$$\mathfrak{B}_0 = (\Omega).$$

Outcome:

failure

because a residual value alone does not specify task relevance, readout, evaluator, admission region, or update rule.

Diagnostic:

missing faithful residual readout.

Branch b_1 : residual with faithful loss readout

Candidate bridge:

$$\mathfrak{B}_1 = (\Omega, \text{Loss}_\tau).$$

If Loss_τ is faithful, residual distinctions become loss distinctions.

Outcome:

partial success

because target distinctions about evaluator outcome and admission are not yet determined.

Diagnostic:

missing evaluator and admission region.

Branch b_2 : faithful loss plus evaluator/admission

Candidate bridge:

$$\mathfrak{B}_2 = (\Omega, \text{Loss}_\tau, E_\tau, W_\tau^{\text{adm}}, X_\tau).$$

If E_τ is induced by the faithful loss and W_τ^{adm} is declared, residual distinctions that cross evaluator or admission thresholds become learning-pressure distinctions.

Outcome:

success for learning pressure.

Diagnostic:

learning pressure is recovered, but physical dynamics is not.

Branch b_3 : physical dynamics bridge

To infer physical dynamics, add coupling data:

$$\mathfrak{B}_3 = (\Omega, \text{Loss}_\tau, E_\tau, W_\tau^{\text{adm}}, X_\tau, \mathcal{A}_{\text{dyn}}),$$

where $\mathcal{A}_{\text{dyn}}$ is an action, control law, field equation, or update rule.

Outcome:

conditional success for dynamics only if $\mathcal{A}_{\text{dyn}}$ is adequate.

This trace shows that residual-to-learning pressure is easier than residual-to-physical dynamics. The failed and partial branches identify why.

23.6 Branch trace II — Positive information geometry to Lorentzian structure

Transfer claim

$$G_+ \Rightarrow G_L.$$

Target distinctions

$$D'_Y = \{\text{indefinite signature, time-like direction, causal cones, conformal structure, metric scale}\}.$$

Branch b_0 : real reparametrization of positive metric

Candidate bridge:

$$\mathfrak{B}_0 = (G_+, \text{real nondegenerate reparametrization}).$$

Outcome:

failure

because positive or positive-semidefinite inertia cannot produce Lorentzian indefinite signature under real nondegenerate reparametrization.

Diagnostic:

missing first order-like, indefinite, analytic, or signature-changing ingredient.

Branch b_1 : analytic / Wick bridge

Candidate bridge:

$$\mathfrak{B}_1 = (G_+, \text{analytic continuation or Wick-type rule}).$$

Outcome:

partial or conditional success

only when the analytic domain, continuation prescription, and interpretation of the continued structure are declared.

Diagnostic:

signature can be changed only by a declared analytic bridge, not by positivity alone.

Branch b_2 : causal order bridge

Candidate bridge:

$$\mathfrak{B}_2 = (G_+, \preceq_{\text{caus}}).$$

Outcome:

partial success

because causal order can supply conformal causal structure but not necessarily scale or full metric data.

Diagnostic:

missing scale / volume / conformal factor.

Branch b_3 : order plus volume bridge

Candidate bridge:

$$\mathfrak{B}_3 = (G_+, \preceq_{\text{caus}}, \text{Vol}).$$

Outcome:

conditional success candidate

if the order-volume data satisfy the reconstruction hypotheses needed for continuum Lorentzian approximation.

Diagnostic:

remaining bridge: continuum/manifold approximation and dynamics.

This trace shows that the positive-metric obstruction is not the end of the road. It identifies the first missing bridge ingredient and then classifies candidate repairs by what target distinctions they recover.

23.7 Branch trace III — Local compatibility to global objecthood

Transfer claim

local compatibility $\Rightarrow$ global objecthood.

Target distinctions

$D'_Y = \{\text{compatible family exists, local fixation vs. objecthood, global object assignment}\}.$

Branch b_0 : local proper organizations only

Candidate bridge:

$$\mathfrak{B}_0 = (\{\text{Org}_{\theta_c}^\circ(\mathcal{C})\}_c).$$

Outcome:

failure

because local proper organization does not determine gluing compatibility.

Diagnostic:

missing transition/restriction maps.

Branch b_1 : local organizations plus transport maps

Candidate bridge:

$$\mathfrak{B}_1 = (\text{Ctx}_{\mathfrak{A}}, \mathcal{F}_{\text{Org}}, \rho_{\mathcal{CD}}).$$

Outcome:

partial success

because objecthood is now well-typed as a compatible-family question.

Diagnostic:

need to check gluing / holonomy.

Branch b_2 : flat gluing

If all transition composites are flat on the relevant local organizations, then compatible families exist by transport from a base chart.

Outcome:

success

for the declared objecthood target.

Branch b_3 : fixed-point-free holonomy

If a cycle γ induces:

$$H_\gamma(O) \neq O$$

for every candidate O , then no compatible object family exists.

Outcome:

failure

Diagnostic:

missing flatness or fixed-point-compatible gluing bridge.

This trace shows that local compatibility is not global objecthood; the missing relation is the gluing/transport relation.

23.8 Compressed branch table for the remaining diagnostic cases

Transfer	First failing branch	Candidate repair	Diagnostic next step
Entropy $\rightarrow$ geometry	entropy values alone do not encode all metric/causal distinctions	dictionary, area law, code subspace, order/scale bridge	specify which geometric distinctions are claimed and which bridge recovers each
Recoverability $\rightarrow$ causality	prediction/correlation does not encode intervention direction	structural causal model, intervention semantics	declare intervention class and counterfactual semantics
Prediction $\rightarrow$ explanation	predictive sufficiency does not encode explanatory distinctions	interpretability task, causal bridge, faithful explanation loss	declare target explanation distinctions and test predictive representation against them

Transfer	First failing branch	Candidate repair	Diagnostic next step
Computation $\rightarrow$ physical law	computational transition does not encode en- ergy/covariance/conservation	physical realization and dynamical law bridge	declare physical implementation constraints

23.9 Branch trace lesson

A bridge-completion trace does not merely say:

success or failure.

It says:

success for which target distinctions, under which bridge, with which remaining failures?

Thus the trace prevents all-or-nothing verdicts. A branch can fail for Lorentzian scale while succeeding for causal order. A branch can succeed for learning pressure while failing for physical dynamics. A branch can make objecthood well-typed while failing by holonomy.

That is the constructive content of the No-Undeclared-Relation principle.

24. Lessons from the worked attempts

The four worked attempts and three full branch traces show different bridge-calibration outcomes.

Transfer	Outcome	Missing or successful bridge
residual $\rightarrow$ learning pressure	success if faithful readout exists; failure if readout collapses residual distinctions	faithful residual-to-loss bridge
local compatibility $\rightarrow$ objecthood	success if gluing exists; failure under fixed-point-free holonomy	flat/fixed-point gluing bridge
positive information geometry $\rightarrow$ Lorentzian	failure without indefinite/order bridge; conditional success with such bridge	causal/order/indefinite/signature bridge
persistence margin $\rightarrow$ trace stability	success below margin; failure at or beyond support boundary	persistence-compatible threshold bridge

These examples close the loop:

question $\rightarrow$ no-free-transfer answer $\rightarrow$ bridge audit $\rightarrow$ bridge calibration $\rightarrow$ success or failure diagnosis.

Both success and failure are informative. Success identifies adequate bridge data. Failure identifies the exact target distinctions still not preserved.

25. Claims and non-claims

25.1 Claims

This paper claims:

1. every cross-context inference is a relation claim and therefore requires a declared relation;
2. every nontrivial cross-context transfer has a source–bridge–target triadic form;
3. the ordinary triangle inequality is a metric/path-cost realization of the more primitive source–bridge–target mediation constraint;
4. nontrivial cross-context inference requires bridge data unless the target statement is boundary/definitional;
5. bridge declarations recover target distinctions from source distinctions;
6. bridge adequacy is the cross-context analogue of task-visible quotient admissibility;
7. bridge declarations can be compared by strength;
8. bridge inferences compose only when their bridge data compose;
9. bridge audits are charted analyses of transfer claims;
10. a bridge is a relation-class, path, trace, resolution, and closure bound depending on level of description;
11. solved bridges can provide candidate templates for underdeclared transfer claims;
12. template transfer is candidate generation, not proof;
13. failed bridge calibration is valid learning;
14. several common foundations transfers are clarified by identifying missing bridge types;
15. constructive bridge completion is a practical application of the No-Free-Transfer theorem;
16. branch-trace analysis distinguishes success, partial success, failure, wrong-direction relations, ill-typed comparisons, and underdeclared bridges;
17. declared relation requires boundary, mediation, and observer/readout faces;
18. undeclared boundary and undeclared observer/readout are NUR failures, not independent rival principles.

25.2 Non-claims

This paper does not claim:

1. no unconditional theorem exists inside a declared context;
2. every foundational derivation is invalid;
3. all external theories are wrong;

4. solved bridges automatically transfer to new domains;
5. entropy cannot contribute to geometry;
6. recoverability cannot contribute to causal reconstruction;
7. positive information geometry cannot participate in Lorentzian emergence;
8. residuals cannot drive dynamics;
9. prediction cannot support explanation;
10. the case studies are exhaustive;
11. bridge-completion supplies empirical predictions by itself;
12. every diagnostic failure is final rather than repairable;
13. boundary or observer/readout are primitives independent of relation;
14. a bare source–target dyad becomes valid without triadic declaration.

The claim is narrower:

nontrivial transfer requires declared preservation structure, and missing bridges can often be constructively class

26. Summary

The master principle is:

No declared relation, no legitimate relational inference.

The No-Free-Transfer theorem is the cross-context corollary:

No nontrivial target distinction transfers across closure contexts without encoding or bridge declaration.

The constructive bridge audit method asks:

What bridge preserves the target distinctions?

Equivalently, every worked example asks how the underdeclared dyad $X \rightarrow Y$ becomes the declared triad $X - \mathfrak{B}_{X \rightarrow Y} - Y$.

If the bridge is missing, one can classify the missing bridge type and search for solved bridge templates in mature theories. A solved template does not automatically transfer. Branch-trace analysis then records which candidate bridges succeed, fail, or partially succeed, and what each result teaches. It becomes a candidate bridge completion that must be calibrated against the target distinctions of the new context.

Thus the no-go is also a method:

find the missing bridge type, then look for a solved bridge of that type.

If the template succeeds, the underdeclared claim becomes inspectable. If it fails, the failure identifies which target distinctions remain unpreserved.

In this sense, both successful and failed bridge completions are learning.

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